Project Reflections - Achievement 2

Predicting Influenza Burden to Support Healthcare Staffing Decisions in the United States of America (2009–2017)

By Bhavana Sawant

Exercise 2.9 – Storytelling with Data Presentations

Tableau link: [Predicting Influenza Burden to Support Healthcare ?Staffing Decisions in the United States (2009–2017) | Tableau Public]

Introduction

Seasonal influenza continues to pose a significant challenge to healthcare systems across the United States. Each year, hospitals and clinics experience predictable surges in patient volume, placing strain on medical staff and resources. For healthcare staffing agencies, the ability to anticipate where and when these surges will occur is critical to ensuring continuity of care and protecting vulnerable populations.

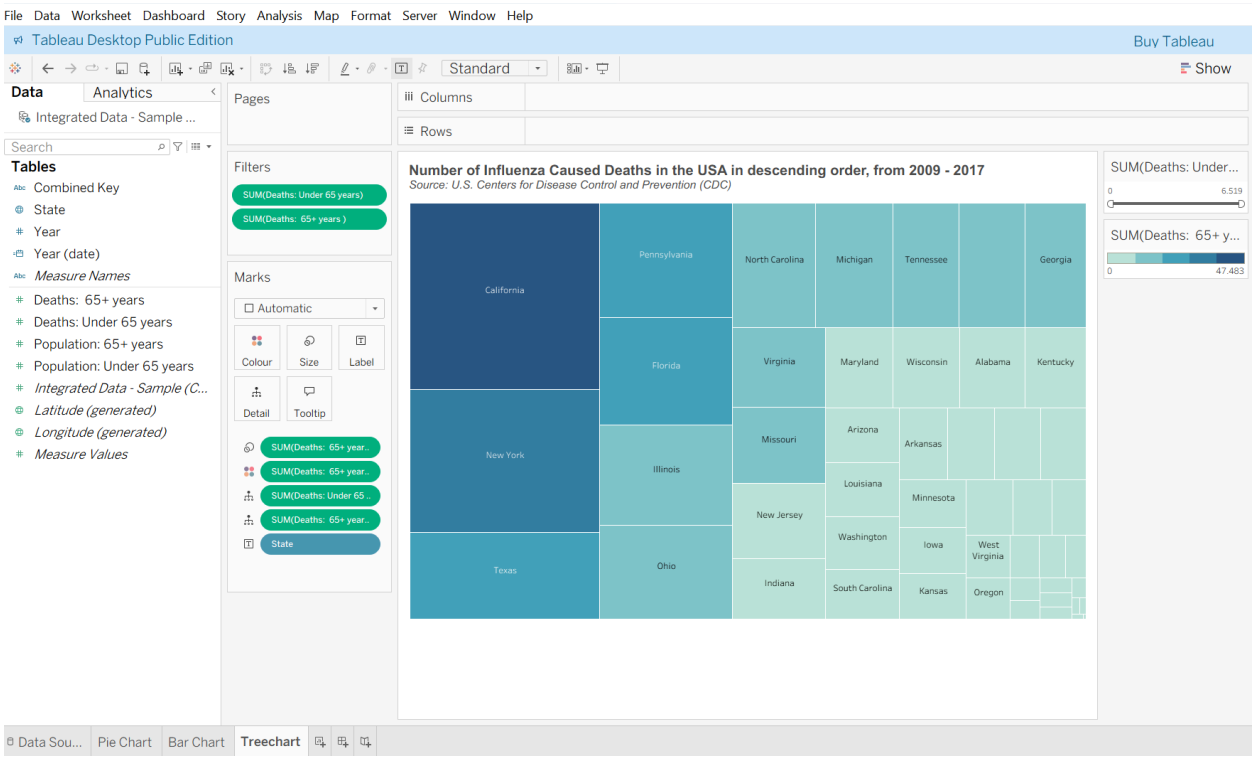
This project analyzes influenza-caused mortality data from 2009 to 2017 to support proactive staffing decisions. By examining geographic patterns, age-based risk, and seasonal trends, the analysis aims to identify areas of consistently high burden and periods of increased demand. The findings provide evidence-based insights that staffing agencies can use to allocate resources more effectively.

Geographic Distribution of Influenza-Caused Deaths

An initial examination of total influenza-caused deaths by state reveals a clear imbalance in disease burden. The treemap visualization shows that a small number of states contribute a disproportionately large share of total deaths over the study period. These high-burden states represent regions where healthcare systems are consistently under greater pressure during influenza seasons.

This uneven distribution highlights the importance of regional staffing strategies rather than uniform national approaches. States with historically high influenza mortality may require

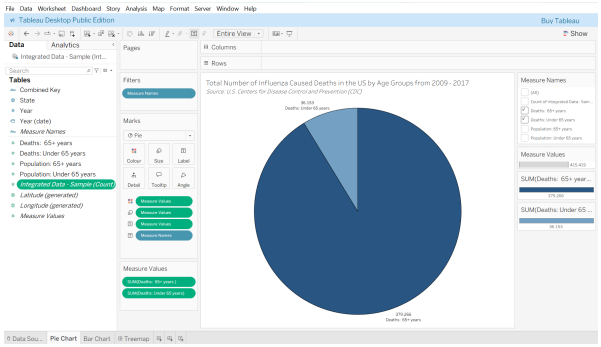
sustained staffing support, while lower-burden states may benefit from more flexible or seasonal resource allocation.

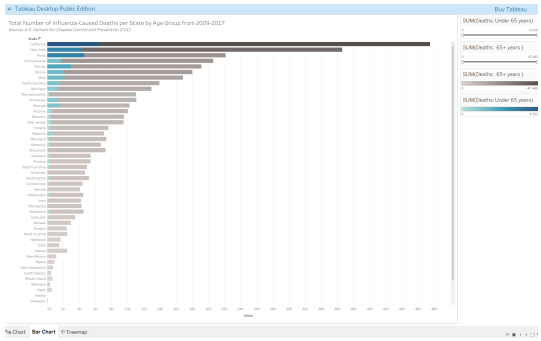


Age-Based Vulnerability

A breakdown of influenza-caused deaths by age group further clarifies where risk is most concentrated. Across nearly all states, individuals aged 65 and older account for the majority of influenza-related deaths. This pattern is consistent over time and across geographic regions.

The dominance of the 65+ age group underscores the vulnerability of older adults to severe influenza outcomes. From a staffing perspective, this suggests a heightened need for clinicians experienced in geriatric care, respiratory treatment, and chronic condition management, particularly in areas with large elderly populations.

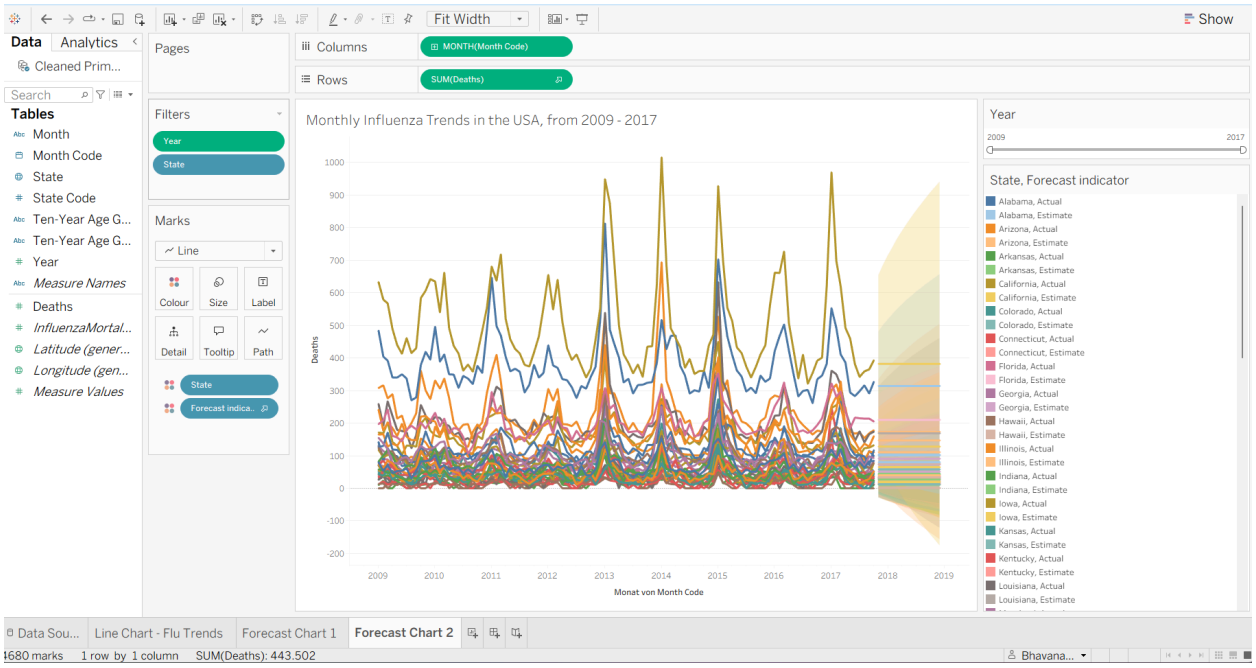




Temporal Patterns and Seasonality

Analysis of monthly influenza trends reveals a strong and recurring seasonal pattern. Influenza activity rises sharply during fall and winter months before declining in the spring and summer. Forecasting models confirm that these peaks are highly predictable.

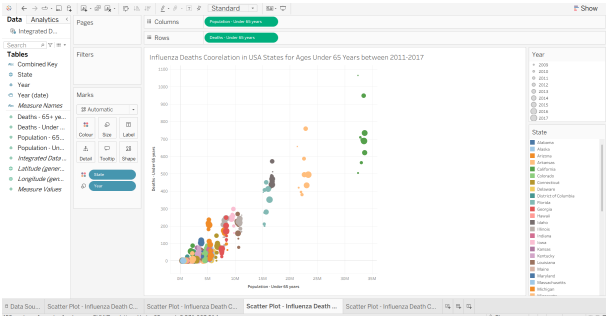
The presence of consistent seasonal cycles presents an opportunity for proactive staffing. Rather than responding to shortages after patient volumes rise, healthcare organizations can plan staffing increases ahead of anticipated peaks. This proactive approach can reduce burnout, improve patient outcomes, and stabilize healthcare delivery during high-demand periods.

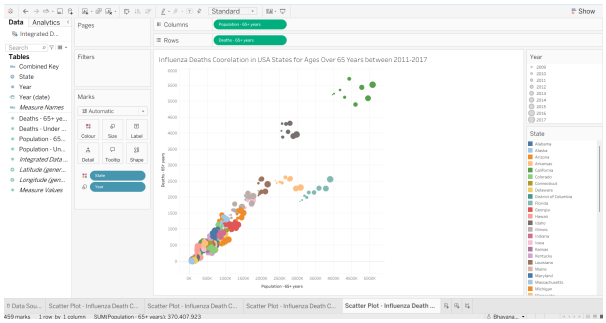


Correlation Between Age Groups

The scatter plot examining the relationship between influenza deaths among populations under and over age 65 reveals a positive correlation. Severe influenza seasons affect all age groups; however, deaths among the 65+ population increase at a significantly faster rate.

This finding reinforces the importance of focusing staffing resources on services that support older adults during severe seasons. While overall healthcare demand rises, the burden placed on facilities serving elderly patients increases disproportionately.

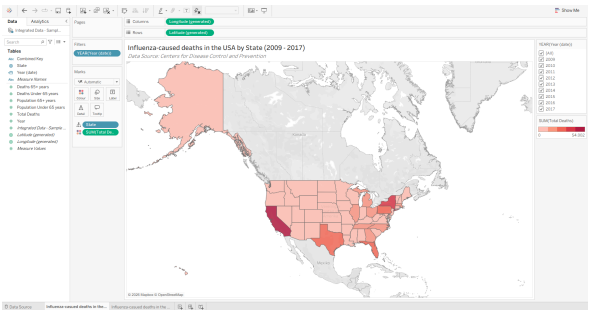


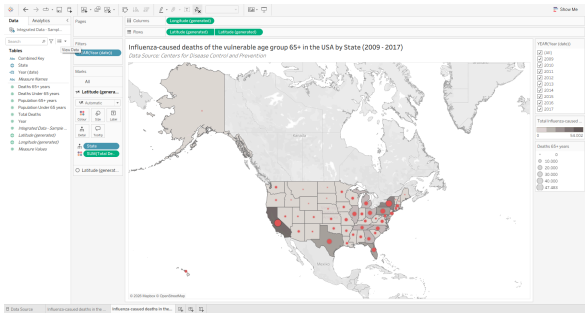


Spatial Risk Among Older Adults

Mapping influenza-caused deaths among adults aged 65 and older further highlights persistent geographic risk. Certain states consistently show higher mortality levels across the study period. These spatial patterns remain relatively stable, suggesting long-term structural or demographic factors rather than isolated outbreaks.

For staffing agencies, this consistency supports the development of long-term regional staffing partnerships rather than temporary or reactive deployments.





Conclusions and Recommendations

This analysis demonstrates that influenza-caused mortality in the United States follows clear geographic, demographic, and temporal patterns. Older adults are consistently the most

vulnerable population, influenza activity is highly seasonal, and certain states experience persistently higher mortality.

Based on these findings, healthcare staffing agencies should prioritize proactive planning, focusing on high-burden states and increasing staffing levels ahead of predictable seasonal peaks. Special attention should be given to facilities serving large elderly populations, where demand is likely to rise most sharply during severe influenza seasons.

Future work could incorporate hospital capacity data, vaccination rates, and population-adjusted mortality metrics to further refine staffing forecasts and enhance decision-making.

Exercise 2.10 – Presenting Findings to Stakeholders

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Reflections: Data Limitations and Metrics

1. Analysis Limitations and Future Wish List

One limitation of this project is reliance on historical mortality data rather than real-time influenza case or hospitalization data. If hospitalization or ICU admission data were available, the analysis could better estimate staffing needs in terms of workload rather than mortality alone. Additionally, access to vaccination rates by state and age group would allow for a deeper understanding of preventive impacts.

2. Data Quality and Bias Considerations

The influenza mortality data includes records with suppressed or unstated age values, which may slightly underestimate true mortality counts. Census population data is based on estimates and may not perfectly reflect yearly demographic changes. There is also potential reporting bias in death certificate data, as influenza-related deaths may be attributed to secondary complications rather than influenza itself.

3. Monitoring the Impact of Staffing Recommendations

To evaluate the effectiveness of staffing changes, healthcare systems could monitor metrics such as patient wait times, staff-to-patient ratios, emergency department overflow rates, and influenza-related mortality trends year over year. A reduction in peak-season strain indicators would suggest improved staffing alignment.

4. Key Monitoring Metric

A useful metric would be influenza-related hospital admissions per available medical staff during peak months. Tracking this metric over time would help determine whether proactive staffing reduces system strain and improves patient outcomes.